

4.6.10

EE25BTECH11022 - sankeerthan

Question: Find the equation of the line passing through the point $(1, -3, 2)$ and parallel to the line

$$\mathbf{r} = (2 + \lambda)\hat{i} + \lambda\hat{j} + (-1 + \lambda)\hat{k} \quad (0.1)$$

Solution: Given line is

$$\mathbf{r} = \begin{pmatrix} 2 + \lambda \\ \lambda \\ -1 + \lambda \end{pmatrix} \quad \mathbf{a} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \quad (0.2)$$

The vector equation of given line is given by

$$\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (0.3)$$

The direction vectors of given line are

$$\mathbf{m} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (0.4)$$

The lines with direction vectors $\mathbf{m}$ and $\mathbf{n}$ are parallel if

$$\mathbf{m} = \mathbf{n} \implies \mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (0.5)$$

The equation of a line is given by

$$\mathbf{r} = \mathbf{a} + t\mathbf{n} \quad (0.6)$$

$$\mathbf{r} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (0.7)$$

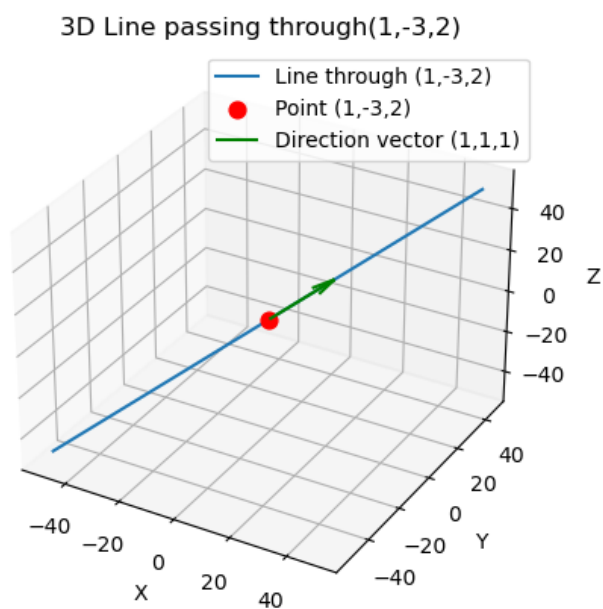


Fig. 0.1